Analysis of LLM Brainstorming Dialogues using the Polyphonic Model

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polyphony-analyzer

1 Objective

This paper offers a comparative analysis of three modes of interaction in terms of brainstorming based on the polyphonic dialogue model. The selected patterns of interaction are as follows: chatbot-chatbot (Optimist-Skeptic), chatbot-human interaction, and human-human interaction between the authors (Filip and Justin).

The analyzed sessions are comprised of short dialogues of 14-18 utterances in length. The objective of the analysis is to determine the theme-based voices and distinguish between divergence as disagreement or reframe and convergence as agreement or synthesis. The chosen dialogues revolve around issues associated with artificial intelligence. They include debates regarding the future timelines of artificial general intelligence development and ethics/legal questions concerning medical artificial intelligence systems.

2 Methodology and Computational Logic

The presented results are obtained through a structured four step analytical pipeline based on the Bakhtinian Polyphonic Model:

- **Utterance Parsing** The conversational turns are parsed for the speaker's identity and the text itself. The text is preprocessed by filtering out stop words, retaining semantically significant words necessary for detecting topics and meaningful conceptual structures.
- **Relation Mapping** A conversation graph is generated by establishing connections between utterances using explicit referential links or semantic relatedness. Semantic relatedness is calculated through TF IDF embeddings in a limited lookback interval. In case no relationship is found, the utterance is connected with its preceding utterance to preserve conversational continuity.
- **Thematic Voice Extraction** Thematic voices refer to repetitive themes that arise from the interactions of the participants regarding common themes. A theme can be classified as a voice if it occurs frequently enough and if many people use it. The program checks for the frequency of the theme and its occurrence among different people across the dialogue.
- **Interanimation Classification** The computer program analyzes the relationships between different messages and decides whether they have diverging or converging ideas. Discourse markers are preferred if they exist. Other markers, such as concession statements, can be classified as convergent messages as well in nuanced conversational contexts.

3 Chatbot-Chatbot Session

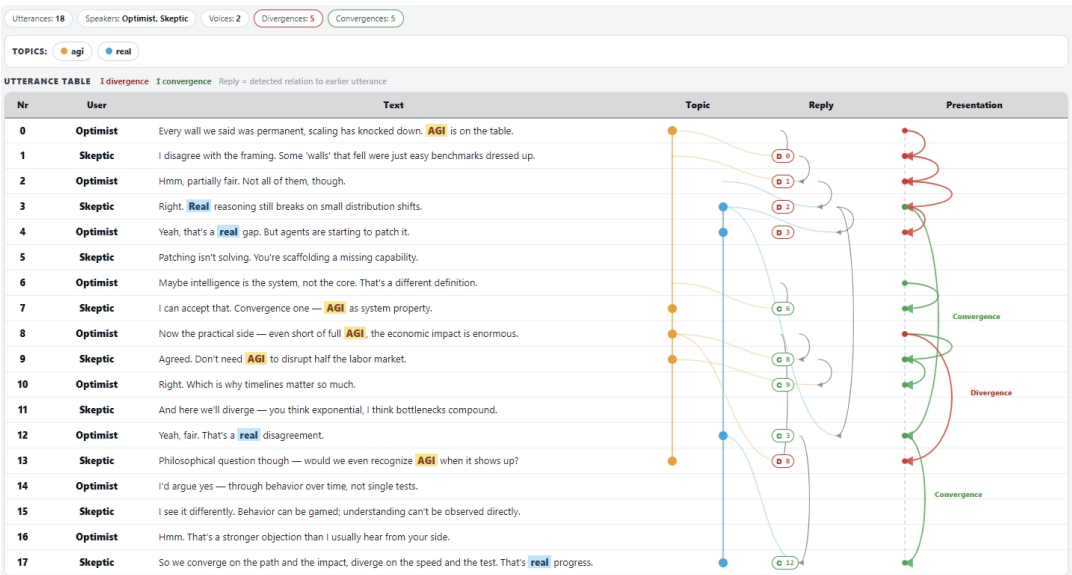

Figure 1: Chatbot-Chatbot polyphonic analysis

The interaction between the Optimist and the Skeptic is characterized by a high level of structure in the argumentative process. In the conversation, the number of points of difference and convergence is identical, suggesting that there is a balanced flow of conflicting ideas and reconciliation.

The prominent speakers include *agi* and *reasoning*. They question each other's presumptions and systematically develop their points. The argument starts from a point of divergence over scale and ends up with agreement on economic consequences.

4 Human-Chatbot Session

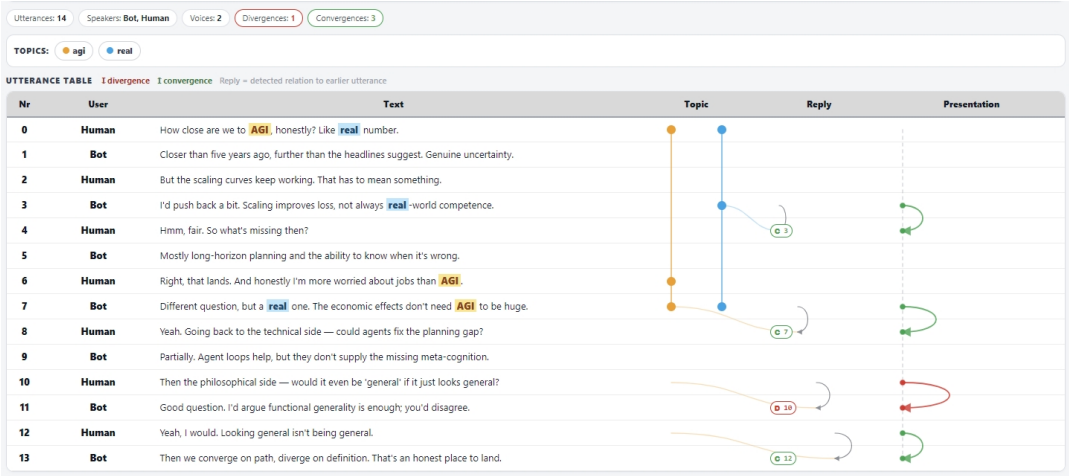

Figure 2: Human-Chatbot polyphonic analysis

The human-chatbot interaction, however, is linear in nature. This is because the human asks direct questions, and the chatbot gives explanations that are relevant to those specific questions. There is less divergence and more convergence, which shows that there was little or no conflict at all in this interaction. The theme revolves around planning abilities and meta-cognition.

5 Human-Human Session

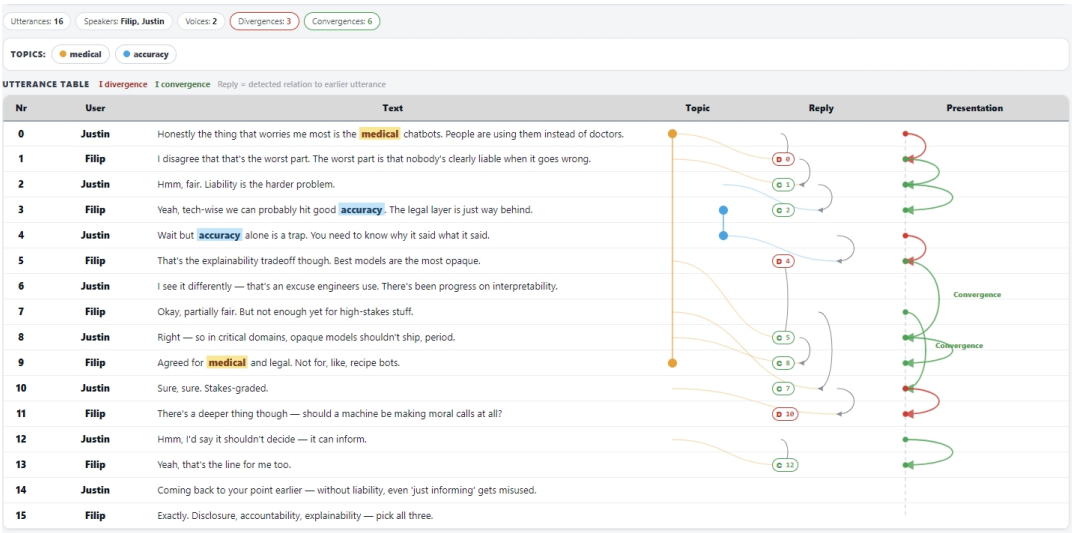

Figure 3: Human-Human polyphonic analysis

The interaction between humans is more organic and contextual. The debate is centered around medical AI, accuracy, and ethical considerations.

Differences naturally emerge based on the participants' different interests, especially in matters of liability and robustness of the AI systems. Nevertheless, the debate begins to converge on the necessity for transparency and accountability of AI in sensitive applications.

6 Comparative Results

Setting	Main Voices	Divergence / Convergence	Interaction Characteristics
Chatbot-Chatbot	agi, reasoning	5 / 5	Balanced structure, explicit argumentation, iterative refinement
Human-Chatbot	agi, planning	1 / 3	Question driven, guided clarification, low conflict
Human-Human	medical, accuracy	3 / 6	Context grounded, ethical reasoning, progressive alignment

Table 1: Comparison of dialogue dynamics

In terms of structural symmetry, the Chatbot-Chatbot configuration shows the greatest symmetry. Both of the two interactants argue in this case, which leads to a very balanced situation regarding divergences and convergences. There is a stepwise progression involving first divergences and then convergences.

The Human-Chatbot configuration is quite asymmetrical. Human participants initiate interactions while chatbots provide structured responses, which is why there are fewer divergences and a greater number of convergences. Such an interaction resembles teaching.

The Human-Human configuration corresponds to the normal course of human conversation. In this case, divergences are less structured and are caused by differences in perspectives that depend on the context. Convergences result from negotiations.

7 System Architecture

The designed architecture uses a modular design to segregate data processing, analysis, and reporting functions. The heart of the system is the analysis engine, which converts raw conversation data into an organized form of utterances, voices, and interactions.

The pipeline consists of the following layers:

- **Input layer** dealing with the processing of text data from the conversations.
- **Processing layer** dealing with tokenizing, removing stop words, and normalizing.
- **Analysis layer** dealing with relation mapping, extracting voices, and interanimation classification.
- **Output layer** outputting structured information that can be exported in textual form or as JSON objects.

Offered separation in this architecture setup allows each component to be independently modified or extended without affecting the overall system.

8 Technical Components Overview

Component	**Description**	**Key Elements**
Data Structures	Structured models used to represent conversational elements such as utterances, voices, and interanimation events.	Utterance: speaker, text, tokens, reply link Voice: label, frequency, speakers, indices Event: type, source, target, trigger, voices
Voice Scoring	Extraction of thematic voices using a combined heuristic based on repetition and distribution across speakers.	Frequency threshold Speaker diversity Collaborative relevance filtering
Relation Detection	Construction of a directed conversation graph linking utterances through explicit references or semantic similarity.	Explicit links (reply) Implicit links (TF IDF + cosine) Fallback to previous utterance
Graph Construction	Representation of the conversation as a connected graph where nodes are utterances and edges encode relationships.	Directed edges Lookback window constraint Continuity enforcement
Interanimation Classification	Rule-based classification of relationships between utterances into divergence or convergence categories.	Disagreement phrases Agreement openers Concession patterns Discourse markers
System Configuration	Centralized parameter control enabling flexible tuning of the analysis pipeline.	Stop word lists Similarity thresholds Voice frequency thresholds Discourse marker sets

Table 2: Summary of technical components and processing logic

9 Limitations and Observations

Despite its effectiveness, the system has several limitations:

- Voice extraction on surface lexical patterns, which may not capture deeper semantic relations. Second, relation classification depends on predefined discourse markers, which may miss implicit or subtle conversational cues.
- Reliance on TF IDF similarity limits the ability to detect long-range or highly abstract relationships between utterances. These limitations suggest that future improvements could include semantic embeddings or learning-based approaches for relation classification.

10 Conclusion

The Polyphonic Model is very useful in representing the inner workings of many-speaker interactions in terms of voices and interactions.

Chatbot interactions have to be more rational and balanced in terms of oppositions. Human-Chatbot interactions allow for exploratory discussions. Human-human interactions require contextual reasoning and value alignment.

This demonstrates that different types of interaction lead to different types of meaning creation and divergence-convergence balances.